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Enhanced flow boiling heat transfer in embedded hybrid distributed jet/pin-fin microchannel heat sink

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ABSTRACT

An embedded hybrid distributed jet/pin–fin microchannel (JPM) heat sink with flow boiling visualization is constructed in the silicon base to enhance the chip-level heat dissipation. The flow boiling heat transfer characteristics of deionized water in JPM heat sink with jet Reynolds numbers (Re_j) of 244 ~ 732 and inlet subcoolings (ΔT_{sub}) of 20°C ~ 60°C are experimentally investigated and compared with those in distributed jet/smooth microchannel (JSM) heat sink. Two-phase flow patterns in JPM and JSM are simultaneously captured by a high-speed microscope camera. It is found that compared with JSM, critical heat fluxes for JPM are significantly enhanced by 27.7 %~70.8 % due to the effective prevention of reverse flow and local dry-out. Specifically, JPM achieves an extremely high critical heat flux of 1098 W/cm^2 at a small pressure drop of 4.2 kPa when Re_j = 732 and ΔT_{sub} = 40°C. Moreover, JPM can increase the heat transfer coefficient by 33.4 %~51.6 % and decrease the effective thermal resistance by 22.8 %~32.1 % due to more nucleation sites and larger heat transfer surfaces existing in JPM than in JSM. Meanwhile, better flow boiling stability and base temperature uniformity are obtained for JPM because its pin–fin structures can enhance the flow disturbance, promote the phase uniform distribution, and prevent the reverse flow. Particularly note that, although the enhancement in heat transfer is at the cost of the increase in pressure drop, JPM has superior comprehensive thermal–hydraulic performance than JSM, with *PEC*s for JPM compared to JSM being 1.18 ~ 1.29 under different conditions. This study provides a more efficient embedded two-phase electronic cooling scheme by combining pin–fin microchannel with distributed jet impingement.

1. Introduction

The continuous improvement of integration and miniaturization of high-power electronic devices makes the heat flux extremely high, which brings great difficulties to thermal management [1]. For wide band-gap (WBG) devices [2,3], next-generation laser devices [4], and phased array radars [4], etc, the high heat flux over 1000 W/cm^2 at chip level needs to be dissipated. Traditional remote cooling [5,6] (or indirect cooling, as shown in Fig. 1(a)) has difficulties in dissipating such enormous heat flux due to the existence of multiple thermal resistances between the heat source and heat sink, including bulk thermal resistances of silicon substrate, heat spreader, and thermal interface material (TIM)), and contact thermal resistances between neighboring bulk materials. By directly integrating the heat sink into the silicon substrate, embedded cooling [5–8] (or direct cooling, as shown in Fig. 1(b)) can greatly decrease the overall effective thermal resistances by reducing the bulk thermal resistances and eliminating the contact thermal resistances compared to remote cooling. Therefore, embedded cooling has attracted more and more interest in the field of high-power electronic cooling in recent years [9–21].

Microchannel boiling is one of the most common embedded cooling schemes because it is easy to integrate into the chip [22,23]. It also has the advantages of high specific surface area [7,23], high heat transfer coefficient (HTC) [24], and coolant saving [7,25]. However, the size limitation makes flow boiling instability easy to occur in the microchannel. Usually, flow boiling instability causes severe temperature and pressure oscillations, and finally leads to the critical boiling crisis [26–28]. In addition, microchannel boiling can result in high pressure drop due to the large length-diameter ratio [23] and poor temperature uniformity due to the increasing wall temperature along the flow direction [24]. Attempts have been made to overcome the above disadvantages and enhance heat transfer within the microchannel, such as pin–fin microchannel [14–17] and manifold microchannel [18,19,29]. The pin–fin microchannel with capillary effect can enhance heat transfer and improve the flow boiling stability due to the increase in heat transfer

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Nomenclature

A	area, cm^2
D_h	hydraulic diameter, mm
h	heat transfer coefficient, $W/(m^2 \cdot K)$
I	current, A
p	pressure, kPa
Δp	pressure drop, kPa
PEC	performance evaluation criterion
q_{eff}	effective heat flux, W/cm^2
Q_{eff}	effective heating power, W
Q_{loss}	heat lost, W
r_{conv}	convective thermal resistance, $K \cdot m^2/W$
r_{eff}	effective thermal resistance, $K \cdot m^2/W$
Re_j	jet Reynolds number
T	temperature, °C
T_u	temperature uniformity, °C
ΔT_w	wall superheat, °C
ΔT_{sub}	liquid inlet subcooling, °C
u	velocity, m/s
U	voltage, V
V	volume flow rate, m^3/s

Greek symbols

δ	thickness, m
λ	thermal conductivity, $W/(m \cdot K)$
μ	dynamic viscosity, $kg/(m \cdot s)$
ρ	density, kg/m^3

Subscripts

avg	average
b	base
f	fluid
i	RTDs location ($i = 1 \sim 7$)
in	inlet
j	jet
0	referenced heat sink of JSM
out	outlet
Si	silicon
w	wall

Abbreviations

CHF	critical heat flux
DAQ	data acquisition
HTC	heat transfer coefficient
HSC	high-speed camera
JPM	jet/pin–fin microchannel
JSM	jet/smooth microchannel
MEMS	micro-electro-mechanical system
ONB	onset of nucleate boiling
PCB	printed circuit board
PDMS	polydimethylsiloxane
RTDs	resistance temperature detectors

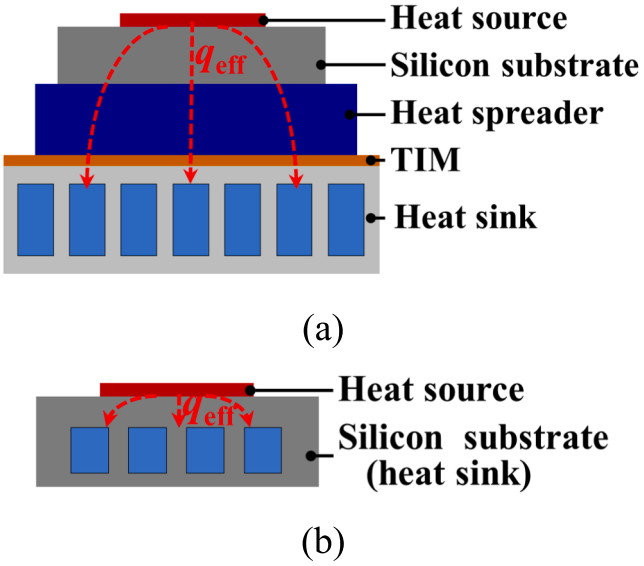


Fig. 1. Schematics of (a) remote cooling, and (b) embedded cooling.

area and nucleation sites [14–17], but it still has the disadvantages of high pressure drop and poor temperature uniformity. The manifold microchannel with shortened effective flow length can dissipate high heat flux and improve temperature uniformity at low pressure drop [18,19,29]. However, it has the challenge of boiling instability. The introduction of pin–fin structure into manifold microchannel may improve its boiling stability, but all the current related studies are numerical simulations [30–32], and experimental studies have not been reported. Moreover, due to the low flow velocity in the manifold microchannel (especially in high-aspect-ratio manifold microchannel), liquid cannot penetrate the vapor layer and re-wet the heating surface in time at high heat flux [19], which makes it easy to occur local dry-out and pre-trigger the critical heat flux (CHF).

Compared with microchannel boiling, jet impingement boiling has advantages in two-phase flow stability, pressure drop consumption, and temperature uniformity [20,21,33–36]. Meanwhile, the HTC of jet impingement can be remarkably high in the stagnation region because of the exceedingly thin boundary layer generated by the intense impact effect [34–36]. However, the HTC decreases rapidly in the regions away from the stagnation region because of the increased thickness of the boundary layer [34–36]. Moreover, jet impingement is not easy to achieve embedded cooling as compared to microchannel [23]. Hybrid jet/microchannel boiling [37,38] combines the advantages of high specific surface area for heat dissipation by microchannel boiling and localized high-intensity cooling by jet impingement boiling, while avoiding the disadvantages of both, and thus has prospects in dissipating high heat flux.

Very few studies on hybrid jet/microchannel boiling could be found and all of them adopted the **remote cooling scheme** (metal-based heat sink) [37–44]. Sung and Mudawar [37–40] conducted some studies on the boiling heat transfer performance in **copper-based single jet and traditional jet/smooth microchannel** heat sinks. They concluded that increasing the jet Reynolds number and inlet subcooling significantly delayed the onset of nucleate boiling (ONB) to higher wall temperature and higher heat flux, but improved CHF. A maximum heat flux (i.e. CHF) of 790 W/cm^2 was obtained at jet Reynolds number (Re_j) of 5314 and inlet subcooling (ΔT_{sub}) of 52°C (i.e. inlet temperature ($T_{f,in}$) of 20 °C (ambient temperature)). Chien et al. [41] compared the boiling heat transfer characteristics of **copper-based traditional jet/smooth microchannel** heat sinks with three jet holes and five jet holes per channel. They concluded that at the same inlet volume flow rate, the thermal resistance of the heat sink with three jet holes per channel was better than that of the heat sink with five jet holes per channel due to the weakening of crossflow interference and the increase of jet velocity. A maximum heat flux of 159 W/cm^2 was obtained at $Re_j = 2443$ and $\Delta T_{sub} = 0$°C (i.e. $T_{f,in} = 30$ °C). Hoang et al. [42,43] investigated the effects of

microchannel height on flow boiling performance in **copper-based traditional jet/smooth microchannel** heat sinks. They discovered that the effective thermal resistance of the heat sink decreased first and then increased with increasing microchannel height. A maximum heat flux of 80 W/cm^2 was achieved at $Re_j = 3406$ and $\Delta T_{sub} = 25°C$ (i.e. $T_{f,in} = 26$ °C). Joshi et al. [44] evaluated the flow boiling performance in an **aluminum-based single jet/two-layer multi-pass branching microchannel** heat sink, and the maximum heat flux of 174 W/cm^2 and maximum HTC of 28.7 kW/(m^2·K) were obtained at $Re_j = 10024$ and $\Delta T_{sub} = 6°C$ (i.e. $T_{f,in} = 55$ °C).

The above literature review indicates that: 1) all the existing hybrid jet/microchannel boiling studies are focused on the metal base, which is suitable for **remote cooling** [5,6]. No experimental study has been reported on **embedded** hybrid jet/microchannel boiling [5,6]. 2) All the existing jet/microchannel boiling schemes apply the **single jet or traditional jet/smooth microchannel** type. The single jet has the disadvantage of poor temperature uniformity, and the traditional jet array (referred to as "traditional jet", for which the outlets are located on the jet chamber's sides) has the disadvantage of crossflow interference, which weakens the heat transfer by decreasing the jet velocity over the heating surface [41,45,46]. However, the distributed jet array (referred to as "distributed jet", for which the outlets are placed near the jet holes for localized discharge of the spent working fluid) can well solve the disadvantages caused by the single jet and traditional jet [46]. Unfortunately, there has been no experimental work on the **distributed jet/microchannel boiling**. 3) Most of the existing studies on jet/microchannel boiling use **smooth microchannel**, but there are few studies on the improvement of microchannel structure. Pin-fin structure (a kind of simple but efficient enhanced heat transfer structure) has been widely used in microchannel [14–17], the research on enhanced boiling heat transfer by introducing pin–fin structure into jet/microchannel is urgently demanded. **Jet/pin–fin microchannel boiling** has great potential yet to be explored, and the internal heat transfer mechanism is still scarce. Thus, it needs to be comprehensively and deeply studied. Particularly note that different from the pin–fin manifold microchannel, the jet/pin–fin microchannel has high jet velocity and strong jet impingement effect. Therefore, the jet/pin–fin microchannel boiling has higher HTC, better boiling stability (effectively preventing reverse flow), and higher heat flux dissipation (re-wetting the heating surface in time to avoid dry-out at high heat flux) than the pin–fin manifold microchannel boiling in theory.

Given the above analysis, to investigate a more effective two-phase cooling scheme at the chip level, an embedded (silicon-based) hybrid distributed jet/pin–fin microchannel (JPM) heat sink with flow boiling visualization is designed and fabricated in this study. The flow boiling experiments of deionized water in JPM heat sink with jet Reynolds numbers of 244 ~ 732 and inlet subcoolings of 20°C ~ 60°C are conducted and compared with those in distributed jet/smooth microchannel (JSM) heat sink. The flow boiling curve, heat transfer coefficient and effective thermal resistance, boiling stability and temperature uniformity, pressure drop and thermal–hydraulic performance for JPM and JSM are comprehensively investigated. Two-phase flow patterns in embedded hybrid distributed jet/microchannel heat sinks are also investigated in detail.

2. Experimental method

2.1. Experimental setup

The schematic plot of the experimental setup for hybrid jet/microchannel boiling is shown in Fig. 2. The degassed and deionized water in the reservoir, being pumped by a gear pump (Micropump, GA-V21), flows through a volume flow meter (Omega, FLR-1604A) and then to a preheater, which is connected with a thermostatic bath. The water is heated by the preheater to a desirable temperature. Following that, the deionized water with a certain inlet temperature enters the test section, in which it is heated using a serpentine film microheater integrated on the bottom of the embedded hybrid distributed jet/pin–fin microchannel heat sink (will be introduced in Section 2.2). To prevent impurities from clogging the equipment in the setup, two filters of sizes 2 μm and 0.5 μm (Swagelok) are placed in front of the pump and test section, respectively. After exiting the test section, the hot deionized water and vapor are condensed by a condenser connected with a thermostatic bath and finally return to the reservoir. A DC power supply (Itech, IT6874A) provides the heating power to the heat sink. A high-speed camera (HSC, Photron, Fastcam Mini AX200, 6400fps) integrated with a microscope lens (Navitar, Zoom 6000) enables the visualization of boiling flow patterns in the distributed jet/pin–fin microchannel heat sink. Two pressure transducers (GE, PMP5023) and two thermocouples (Omega, TT-T-36, diameter of 0.5 mm) are used to measure the fluid inlet/outlet pressures and temperatures through the heat sink, respectively. Seven resistance temperature detectors (RTDs) are used to measure the base temperatures of the heat sink (will be introduced in Section 2.2). After being recorded using a data acquisition (DAQ) system (Agilent, 34972A), all the heating power, temperature, pressure, and flow rate measurements are transmitted to a computer for combination analysis with two-phase flow patterns.

2.2. Test section and hybrid distributed jet/pin–fin microchannel heat sink

The test section used in this paper is given in Fig. 3(a). From top to bottom, the test section (60 × 60 × 31.6 mm^3) consists of a cover plate (thickness of 15 mm), a gasket (thickness of 5 mm), a hybrid distributed

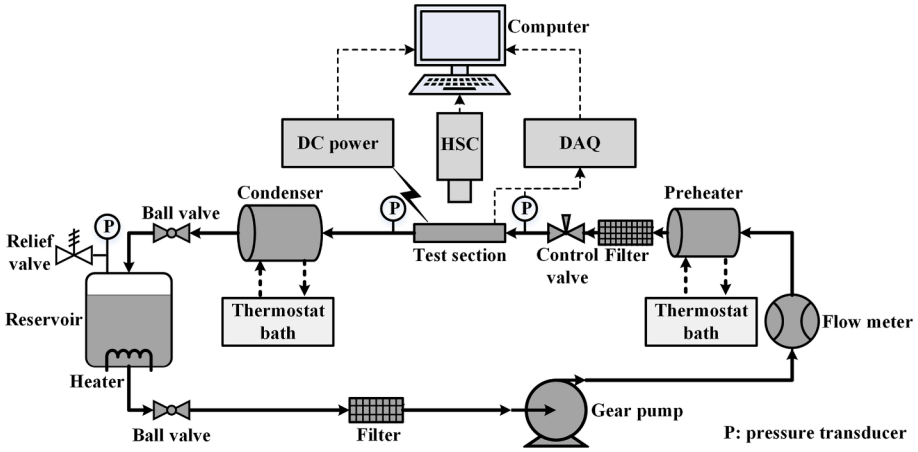


Fig. 2. Schematic plot of the experimental setup.

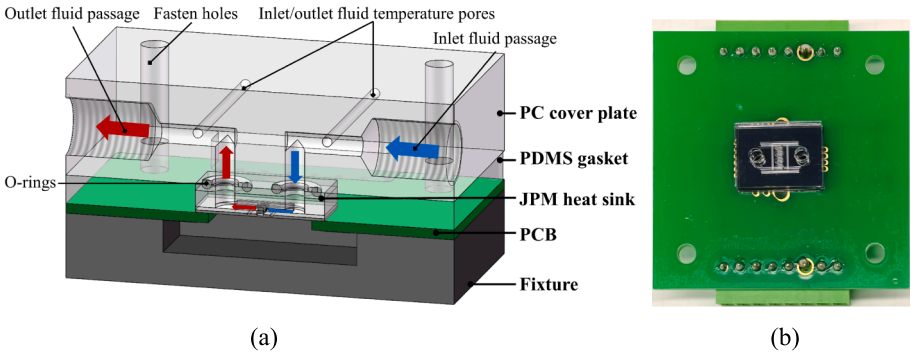


Fig. 3. (a) Drawing of the test section with a half-symmetry section removed, and (b) photograph of the embedded heat sink equipped with PCB.

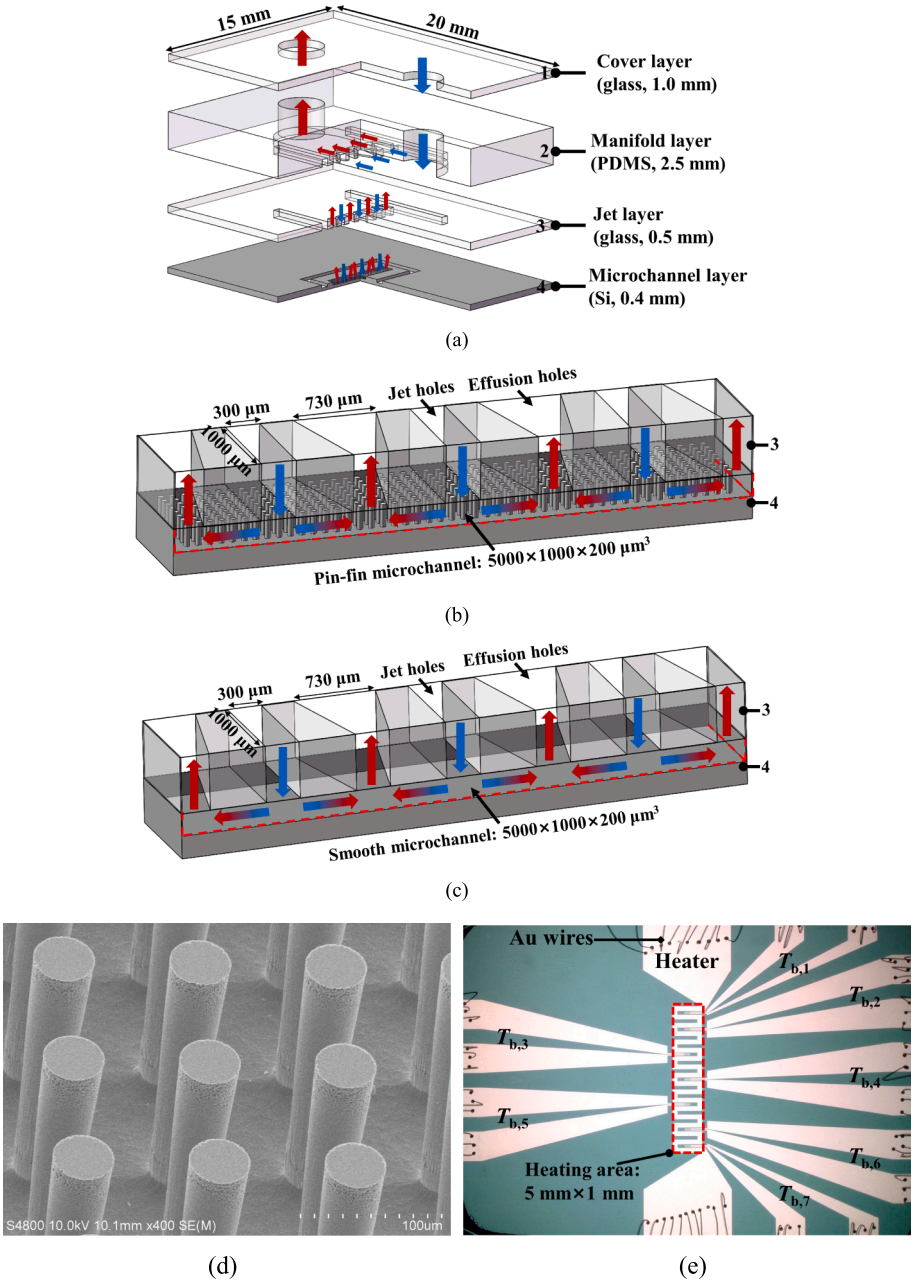


Fig. 4. Embedded hybrid distributed jet/pin–fin microchannel heat sink with flow boiling visualization: (a) exploded view of the heat sink (with a quarter-section removed), (b) and (c) design and details of the jet layer and microchannel layer for JPM and JSM, respectively, (d) SEM image of the pin-fins for JPM, and (e) enlarged view of the microheater and RTDs integrated on the bottom of the heat sink.

jet/pin–fin microchannel (JPM) heat sink, a printed circuit board (PCB, thickness of 1.6 mm), and a fixture (thickness of 10 mm). The inlet/outlet fluid temperature pores and passages are located in the polycarbonate (PC) cover plate. To prevent fluid leakage, two O-rings are inserted between the cover plate and the jet/pin–fin microchannel heat sink. The PCB is connected to the microheater and RTDs of the embedded hybrid distributed jet/pin–fin microchannel heat sink via Aurum (Au) lead wires (diameter of 50 μm). The heat sink equipped with PCB (as shown in Fig. 3(b)) is positioned between the cover plate and the fixture. To prevent physical damage to the embedded heat sink during the assembly of the test section and decrease heat loss from the heat sink, a polydimethylsiloxane (PDMS, thermal conductivity of 0.146 W/(m·K)) gasket surrounding the heat sink is placed between the PCB and the cover plate. After all of the aforementioned components are secured with four bolts, the test section with the hybrid distributed jet/pin–fin microchannel heat sink is acquired.

Fig. 4(a) gives the exploded view of the embedded (silicon-based) hybrid distributed jet/pin–fin microchannel heat sink with flow boiling visualization. The embedded heat sink (20 mm × 15 mm × 4.4 mm) is bonded by four functional structural layers, which are a 1.0-mm-thick cover layer, a 2.5-mm-thick manifold layer, a 0.5-mm-thick jet layer, and a 0.4-mm-thick microchannel layer, from top to bottom. Note that to realize the visualization of boiling flow patterns in the silicon microchannel layer, transparent materials of glass, PDMS, and glass are used in the cover layer, manifold layer, and jet layer, respectively. The cover layer has two holes (diameter of 3 mm) for the inlet and outlet of deionized water. Under the cover player, a "W" type barrier is designed in the manifold layer to isolate the inlet and outlet of deionized water and improve the uniformity of fluid distribution. As shown in Fig. 4(b) and (c), the jet layer is arranged alternately with jet holes and effusion holes. The dimensions of the slot jet hole are 1000 μm × 300 μm, which corresponds to a hydraulic diameter of 462 μm. The distance between each jet is 1600 μm. The silicon microchannel layer, having the microchannel etched on the top as well as the microheater and RTDs integrated on the bottom of the silicon layer, is designed to heat the deionized water and realize jet/microchannel cooling. As shown in Fig. 4(b), a pin–fin microchannel (5000 μm × 1000 μm × 200 μm) is designed on the top of the microchannel layer. For comparison, a smooth microchannel (no pin–fin structures) having the same size as the pin–fin microchannel is also designed (Fig. 4(c)). For the convenience of following discussions, the hybrid distributed "**jet/pin–fin microchannel**" and "**jet/smooth microchannel**" are simply named "**JPM**" and "**JSM**", respectively. The scanning electron microscope (SEM) image of the local in-line circular pin-fins is also provided in Fig. 4(d). The pin–fin diameter, height, and pitch are 50 μm, 200 μm, and 100 μm, respectively. As shown in Fig. 4(e), a serpentine film microheater and seven RTDs with the same thickness of 100 nm are designed on the bottom of the microchannel layer. The microheater, which covers an area of 5 × 1 mm^2, is used to simulate the heat source. To measure the base temperatures of the heat sink, seven RTDs (200 μm × 75 μm) are designed at the locations that correspond to the jet holes and effusion holes. The embedded hybrid distributed jet/microchannel heat sinks with flow boiling visualization are processed using the MEMS (micro-electro-mechanical system) technology. The detailed MEMS fabrication process of the heat sinks as well as the calibration method of RTDs for base temperature measurement can be referred to our previous work [45] and will not be repeated here.

The flow path for the hybrid distributed jet/pin–fin microchannel heat sink is also shown in Fig. 4(a). After flowing into the manifold layer from the cover layer, the deionized water is evenly divided into the jet holes in the jet layer. The deionized water jets into and then travels along the pin–fin microchannel in the microchannel layer, in which the water is heated by the microheater and the jet/microchannel boiling occurs. After flowing through the effusion holes, the boiled water finally leaves the heat sink.

2.3. Experimental procedure

Experiments are conducted at atmospheric pressure with jet Reynolds numbers of 244 ~ 732 (corresponding to volume flow rate of 2.25 × 10^{-7} ~ 6.75 × 10^{-7} m^3/s) and inlet subcoolings of 20°C ~ 60°C. Before conducting the experiments, the deionized water is first charged into the reservoir and then boiled by the heater to eliminate the effect of non-condensable gases. After that, the deionized water is pumped into the test section at a desirable temperature and flow rate. When the fluid temperature and flow rate become stable, the DC power is supplied to the heat sink. Once the base temperatures of the heat sink measured by RTDs reach a steady state, the required experimental data are recorded, which include the base temperatures, fluid inlet/outlet temperatures and pressures, flow rate, voltage, and current. At the same time, the two-phase flow patterns are captured by the high-speed camera for combined analysis with measured data.

2.4. Data reduction

The effective heat flux q_{eff} to the heat sink can be obtained by:

$$q_{eff} = Q_{eff}/A_b \tag{1}$$

where A_b is the area of the microheater as shown in Fig. 4(e); Q_{eff} is the effective heat to the heat sink, which can be obtained by deducting the heat loss Q_{loss} from the input power (input voltage U times current I):

$$Q_{eff} = UI - Q_{loss} \tag{2}$$

The heat loss Q_{loss} from the heat sink to the environment in this study is calibrated using the extensively used void heating method for the flow boiling experiment [47–49]. As shown in Fig. 5, the heat losses related to the average base temperatures for JPM and JSM heat sinks are obtained, and the corresponding prediction equations for the heat losses are provided as follows:

$$Q_{loss} = 0.0249(T_{b,avg} - 23.48)\ (for JPM) \tag{3}$$

$$Q_{loss} = 0.0236(T_{b,avg} - 23.24)\ (for JSM) \tag{4}$$

The average base temperature, $T_{b,avg}$, can be expressed by:

$$T_{b,avg} = \frac{1}{7}\sum_{i=1}^{7} T_{b,i} \tag{5}$$

where $T_{b,i}$ is the base temperatures measured by seven RTDs ($i = 1 \sim 7$).

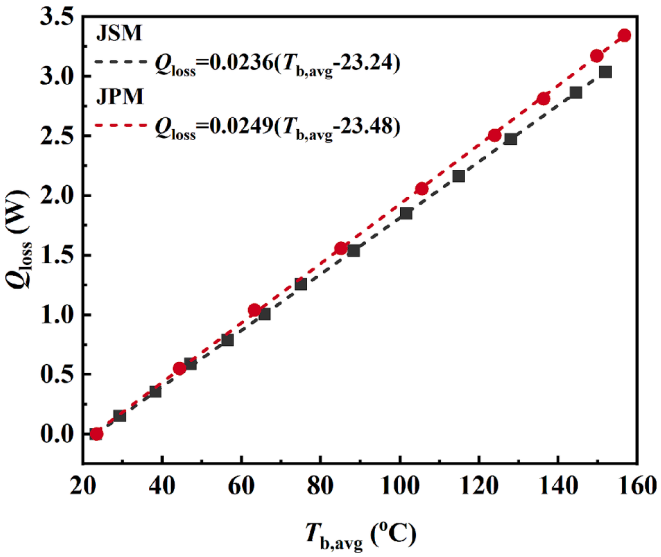


Fig. 5. Heat losses for JPM and JSM heat sinks.

The average heat transfer coefficient, h_{avg}, is obtained using the inlet temperature of the jet fluid, $T_{f,in}$, and is calculated by:

$$h_{avg} = q_{eff}/(T_{w,avg} - T_{f,in}) \tag{6}$$

where $T_{w,avg}$ is the average wall temperature of the microchannel and can be expressed by:

$$T_{w,avg} = T_{b,avg} - q_{eff} \cdot \delta_{Si}/\lambda_{Si} \tag{7}$$

where λ_{si} and δ_{si} are the thermal conductivity and the base thickness of the silicon substrate between the microchannel wall and base, respectively.

The effective thermal resistance per unit area, r_{eff}, is calculated by:

$$r_{eff} = (T_{b,avg} - T_{f,in})/q_{eff} \tag{8}$$

The inlet jet Reynolds number, Re_j, that is, the fluid Reynolds number characterized by the hydraulic diameter of the jet hole, can be calculated by:

$$Re_j = \rho_{in} u_j D_h/\mu_{in} \tag{9}$$

where μ_{in} and ρ_{in} are the inlet fluid's dynamic viscosity and density; D_h is the hydraulic diameter of the jet hole; u_j is the inlet jet velocity, which is calculated by:

$$u_j = V_{in}/A_j \tag{10}$$

where A_j is the area of all three jet holes, and V_{in} is the volume flow rate of water.

The temperature uniformity of the heat sink, T_u, is characterized by the sample standard deviation of base temperatures and can be expressed by:

$$T_u = \sqrt{\frac{1}{6}\sum_{i=1}^{7}(T_{b,i} - T_{b,avg})^2} \tag{11}$$

The pressure drop Δp through the heat sink is obtained by:

$$\Delta p = p_{in} - p_{out} \tag{12}$$

where p_{in} and p_{out} are the inlet and outlet pressures of water, respectively.

To comprehensively evaluate the thermal–hydraulic performance of different heat sinks, the performance evaluation criterion (*PEC*) is adopted [50], which is defined as:

$$PEC = \frac{h_{avg}/h_{avg,0}}{(\Delta p/\Delta p_0)^{1/3}} \tag{13}$$

where h_{avg} and $h_{avg,0}$ represent the average heat transfer coefficients of JPM and JSM (referenced heat sink in this work), respectively; Δp and Δp_0 represent the average pressure drops of JPM and JSM, respectively. Generally, if the *PEC* value of JPM is greater than 1, the overall thermal–hydraulic performance of JPM is better than that of JSM.

2.5. Uncertainty analysis

The uncertainties of the main experimental parameters discussed in this study, including both direct and indirect measurements, are listed in Table 1. The precisions of the measuring instruments are the primary source of uncertainty in direct measurements, such as $T_{b,i}$ ($i = 1 \sim 7$), $T_{f,in}$, $T_{f,out}$, p_{in}, p_{out}, V_{in}, U, and I. The uncertainty propagation analysis method [51] is applied to calculate the uncertainties of indirect parameters, such as q_{eff}, h_{avg}, and r_{eff}. The relative uncertainties can be calculated as:

$$\frac{u(R)}{R} = \frac{1}{R}\sqrt{\left(\frac{\partial R}{\partial v_1}u(v_1)\right)^2 + \left(\frac{\partial R}{\partial v_2}u(v_2)\right)^2 + \cdots + \left(\frac{\partial R}{\partial v_n}u(v_n)\right)^2} \tag{14}$$

Table 1
Uncertainties.

Parameters	Instruments	Uncertainties (±)
$T_{b,i}$ ($i = 1 \sim 7$)	RTDs (calibrated)	0.48 °C
$T_{f,in}$ and $T_{f,out}$	T-type thermocouples (calibrated)	0.30 °C
p_{in} and p_{out}	Pressure transducers	0.12 %
V_{in}	Volume flow meter	1.11 %
U	DC power supply	0.15 %
I	DC power supply	0.49 %
q_{eff}	–	1.33 %
h_{avg}	–	6.77 %
r_{eff}	–	6.14 %

where R represents the indirect parameter which is obtained from the direct measurements such as v_1, v_2, …, v_n, and u represents the uncertainties. As can be shown, the uncertainties of q_{eff}, h_{avg}, and r_{eff} are limited to 1.33 %, 6.77 %, and 6.14 %, respectively.

3. Results and discussion

To reveal the enhanced boiling heat transfer mechanism in hybrid distributed jet/pin–fin microchannel, experiments for two kinds of embedded hybrid distributed jet/microchannel (jet/pin–fin microchannel (JPM) and jet/smooth microchannel (JSM)) heat sinks are carried out in this work. The flow boiling curve, heat transfer coefficient and effective thermal resistance, boiling stability and temperature uniformity, pressure drop and thermal–hydraulic performance of deionized water for JPM and JSM are comprehensively examined and discussed. Two-phase flow patterns in JPM and JSM are also investigated in detail.

3.1. Flow boiling curve

Fig. 6 shows the boiling curves of deionized water for JPM and JSM heat sinks under different test conditions (i.e. jet Reynolds numbers and inlet subcoolings). In the single-phase regime before the onset of nucleate boiling (ONB), the heat flux increases linearly with increasing wall superheat ($\Delta T_{w,avg}$). The slopes of the boiling curves for JPM are higher than those for JSM in the single-phase regime. This is because, on one hand, the addition of micro pin-fins causes the increase in heat transfer area; on the other hand, the pin–fin structures disturb the flow boundary layer, increase the fluid disturbance, and thus enhance heat transfer. In addition, the slope of the boiling curve before ONB increases with increasing jet Reynolds number but remains almost constant with increasing inlet subcooling.

The wall superheat steadily rises with increasing heat flux, reaching ONB and starting the phase change heat transfer. It is found that JPM can pre-trigger ONB compared to JSM, with the incipient wall superheat (i.e. wall superheats at ONB) for JPM being decreased by up to 3.90°C. The pre-triggering of ONB for JPM can be explained by the increase in nucleation sites. Fig. 7 presents the boiling flow patterns in different regions (stagnation region, wall jet region, and recirculation region) at the early boiling stage (after ONB) for JSM and JPM, respectively, under $Re_j = 487$ and $\Delta T_{sub} = 40$°C. Comparing Fig. 7(a) and Fig. 7(b), it can be seen that the isolated bubbles first form in the recirculation region for JSM, while more bubbles form in the recirculation region and wall jet region for JPM. This demonstrates that pin–fin structures can significantly increase the nucleation sites over larger regions. This is because, on one hand, the added surface area of micro pin-fins increases the probability of cavities, thus increasing the opportunities for more nucleation sites; on the other hand, the use of deep reactive ion etching (DRIE, a kind of MEMS technique for etching the microchannel in silicon wafer in this work) makes nucleation sites easier to form on the bottom surface surrounding the pin–fin structures (as shown in the SEM image of Fig. 4(d)). The significant increase in nucleation sites over larger regions is conducive to triggering the incipient boiling. Moreover, the obvious boiling hysteresis phenomenon [45] is observed after ONB for

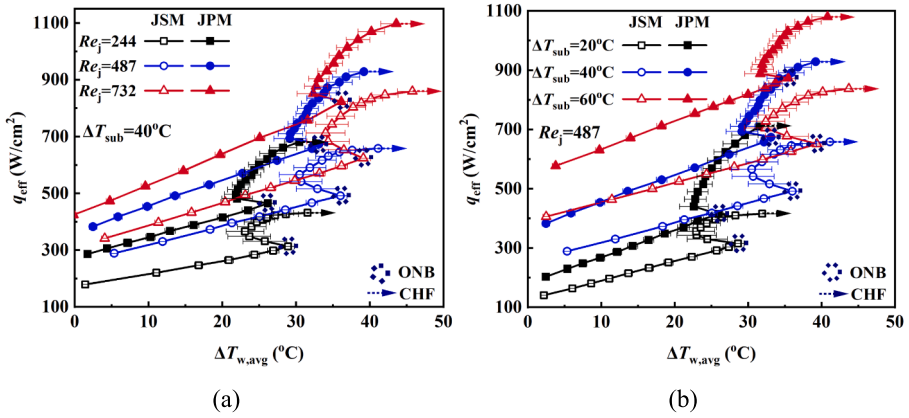


Fig. 6. Boiling curves for JSM and JPM heat sinks under (a) $Re_j = 244 \sim 732$, and (b)$\Delta T_{sub} = 20°C \sim 60°C$.

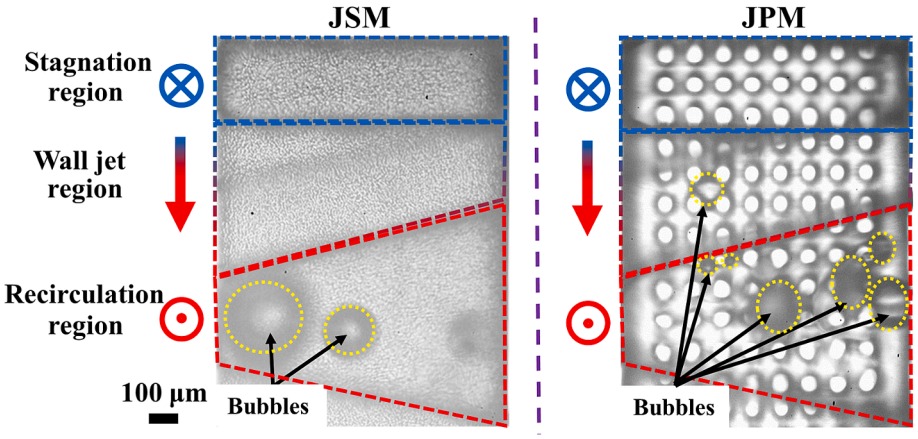


Fig. 7. Boiling flow patterns at the early boiling stage (after ONB) under $Re_j = 487$ and $\Delta T_{sub} = 40°C$ for (a) JSM ($q_{eff} = 516$ W/cm^2) and (b) JPM ($q_{eff} = 693$ W/cm^2).

JSM and JPM. Also note that after ONB, the slopes of the boiling curves for JPM are higher than those for JSM in the two-phase regime due to the increase in heat transfer surface and nucleation sites for JPM.

When heat flux approaches critical heat flux (CHF), the slope of the boiling curve starts to decrease. Compared with JSM, CHFs for JPM are significantly enhanced by 27.7 %~70.8 % under different conditions. Specifically, the embedded hybrid distributed JPM heat sink achieves an extremely high CHF of 1098 W/cm^2 under $Re_j = 732$ and $\Delta T_{sub} = 40°C$. Generally, CHF is triggered because the local dry-out cannot get the wetting of the liquid in time [52]. Fig. 8 presents the boiling flow patterns in different regions at the late boiling stage (before CHF) for JSM and JPM, respectively, under $Re_j = 487$ and $\Delta T_{sub} = 40°C$. At this stage, the stagnation region can produce a large number of bubbles and the coverage area of vapor film in the wall jet region extends to the stagnation region. As shown in Fig. 8(a) for JSM, the rapidly expanding bubble blocks the effusion hole and flows reversely to the jet hole. It is difficult to obtain liquid wetting in time in local dry-out areas, which easily leads to pre-trigger CHF for JSM. However, as shown in Fig. 8(b) for JPM, the stable annular flow with the vapor core dominates the microchannel with the pin–fin structures and leaves a layer of liquid film

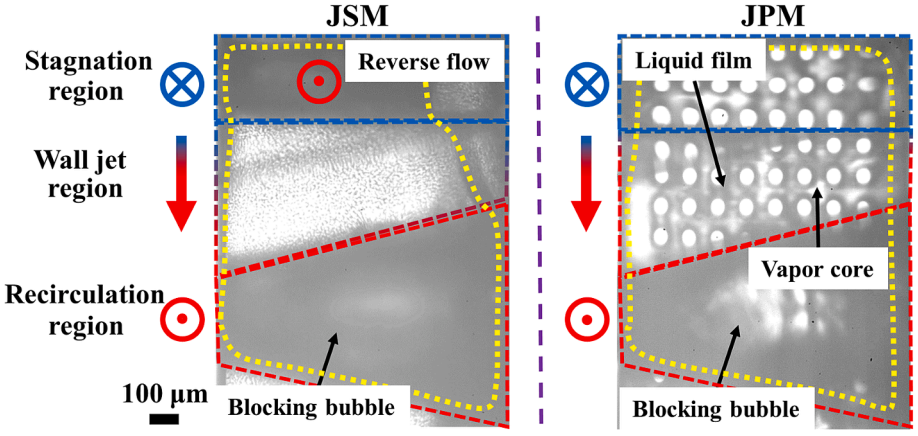


Fig. 8. Boiling flow patterns at the late boiling stage (before CHF) under $Re_j = 487$ and $\Delta T_{sub} = 40°C$ for (a) JSM ($q_{eff} = 652$ W/cm^2) and (b) JPM ($q_{eff} = 910$ W/cm^2).

around the pin-fins, which prevents reverse flow and local dry-out, and thus leads to a higher CHF. In addition, CHF increases with increasing jet Reynolds number due to the vapor layer covering the heating surface can be penetrated more successfully by faster liquid jets, which can delay the occurrence of CHF. CHF also increases with increasing inlet subcooling due to the large condensation effect of the subcooled liquid.

3.2. Heat transfer coefficient and effective thermal resistance

Fig. 9 gives the heat transfer coefficients (HTCs, defined as Eq. (6)) for JSM and JPM heat sinks under different conditions. During the single-phase regime before ONB, HTC is not affected obviously by heat flux. Once boiling occurs, HTC rises significantly with increasing heat flux due to the utilization of latent heat of vaporization. Particularly, an abrupt increase in HTC appears after ONB due to the influence of boiling hysteresis (introduced in Section 3.1). However, HTC decreases when heat flux approaches CHF due to the local dry-out of liquid film. HTCs for JPM are greatly increased compared to those for JSM. In the single-phase regime, HTCs for JPM are higher than those for JSM due to the increase in heat transfer surface and fluid disturbance. In the two-phase regime, JPM undergoes higher boiling heat fluxes (q_{eff}) under the same temperature difference ($T_{w,avg}$-$T_{f,in}$), and the boiling HTCs (q_{eff}/($T_{w,avg}$-$T_{f,in}$)) increase significantly. The main reason for this is the micro pin–fin structures in the microchannel for JPM can produce more nucleation sites than JSM (as explained in Section 3.1). Compared with JSM, the average HTCs for JPM are increased by 33.4 %~51.6 % under different conditions. In particular, the maximum HTC for JPM achieves 137 kW/(m^2·K) under Re_j = 487 and ΔT_{sub} = 20°C. In addition, HTC increases with increasing jet Reynolds number (Fig. 9(a)). However, the inlet subcooling has little influence on HTC in the single-phase regime (Fig. 9 (b)). The heat transfer curve shifts greatly to the left, and its slope in the two-phase regime increases with decreasing inlet subcooling because lower inlet subcooling (i.e. higher inlet fluid temperature) leads to a greater boiling heat transfer intensity at lower heat flux [45].

Fig. 10 gives the effective thermal resistances (r_{eff}, defined as Eq. (8)) for JSM and JPM heat sinks under different conditions. Since wall thermal conductivity and thickness (influencing conduction thermal resistance), as well as fluid HTC (influencing convective thermal resistance) generally remain constant in the single-phase regime for a given operating state, the effective thermal resistance also remains constant. When boiling occurs, the effective thermal resistance decreases significantly due to the increase in boiling heat transfer coefficient. However, when heat flux approaches CHF, the effective thermal resistance increases. A further comparison of Fig. 10 with Fig. 9 shows that the curve trend of effective thermal resistance mirrors the curve trend of HTC under the same condition. It is found that the effective thermal resistance values for JPM are kept significantly lower than those for JSM. This difference can be attributed to the significant increase in HTCs for JPM. Compared with JSM, the average effective thermal resistances for JPM are decreased by 22.8 %~32.1 % under different conditions. The minimum effective thermal resistance for JSM and JPM are 12.9 × 10^{-6} K·m^2/W and 8.7 × 10^{-6} K·m^2/W, respectively, under Re_j = 487 and ΔT_{sub} = 20°C. In addition, the effective thermal resistance decreases with increasing jet Reynolds number (Fig. 10(a)). The HTC increases with increasing jet Reynolds number (Fig. 9(a)), which is a major contributor to the decrease in the effective thermal resistance. Similar to HTC, the effective thermal resistance is not significantly impacted by the inlet subcooling in the single-phase regime (Fig. 10(b)). Nevertheless, the thermal resistance curve shifts greatly to the left with decreasing inlet subcooling in the two-phase regime due to the boiling occurring at lower heat flux.

As discussed above, the effective thermal resistance (r_{eff}) consists of convective thermal resistance (r_{conv} = 1/h_{avg}, influenced by fluid HTC) and conduction thermal resistance (r_{cond} = δ/λ, influenced by the wall thermal conductivity λ and thickness δ of the silicon substrate). To show the contribution of convective thermal resistance to the total effective thermal resistance, Fig. 11 further gives the average ratios of the convective thermal resistance to the effective thermal resistance (r_{conv}/r_{eff}) for JSM and JPM obtained by the present experimental results. It is observed that the convective thermal resistances contribute 88.9 %~93.1 % of the effective thermal resistance for JSM and 85.7 %~89.8 % for JPM under different conditions. The ratios of the convective thermal resistance to the effective thermal resistance for JPM are lower than those for JSM (average reduction of 3.29 %) due to the increase in HTCs.

3.3. Boiling stability and temperature uniformity

As discussed in Section 3.1 (Fig. 8), JSM has a serious reverse flow phenomenon (reverse flow to the jet hole) at the late boiling stage. In actuality, the reverse flow for JSM has appeared at the middle boiling stage. Figs. 12 and 13 show the evolution processes of flow patterns at the middle boiling stage for JSM and JPM, respectively, under Re_j = 487 and ΔT_{sub} = 40°C. As shown in Fig. 12 for JSM, the large bubble generated in the recirculation region rapidly expands to the wall jet region along the reverse direction of incoming flow in a very short time (0.33 ms). However, as shown in Fig. 13 for JPM, the probability of elongated bubbles or slugs expanding in the reverse direction is decreased as they can grow in multiple directions in the wall jet region for JPM. Additionally, the pin–fin structures in the microchannel can enhance flow disturbance, promote uniform distribution of different phases by breaking the bubbles, and trap a small amount of liquid at the intersections of pin-fins. Therefore, in JPM, the prevention of the reverse flow and the uniform distribution of different phases significantly suppress the flow boiling instability occurring and its induced large temperature and pressure variations in JSM. Fig. 14 presents the transient

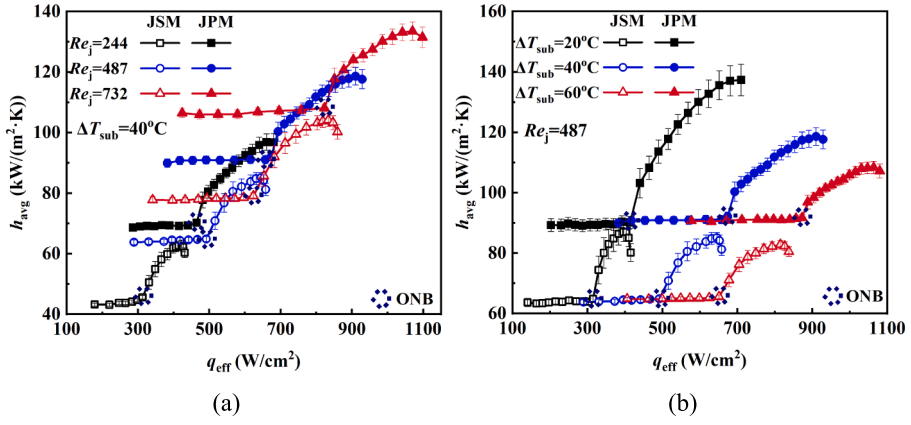


Fig. 9. HTCs for JSM and JPM heat sinks under (a) Re_j = 244 ~ 732, and (b) ΔT_{sub} = 20°C ~ 60°C.

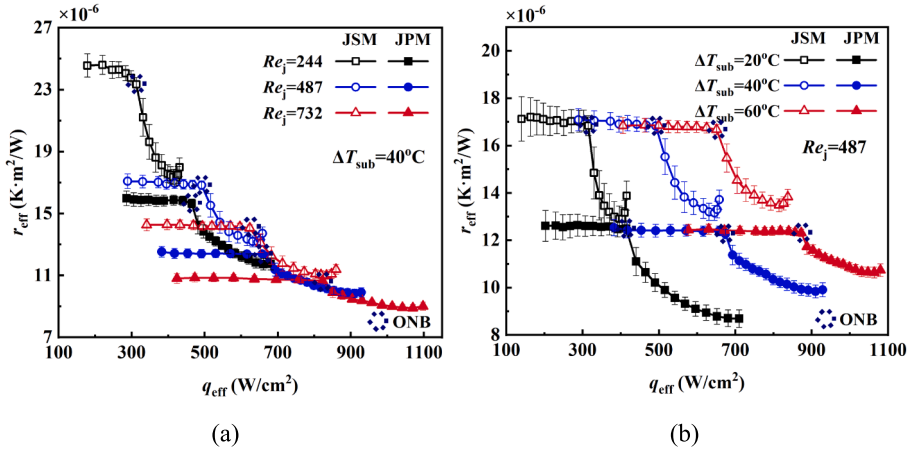


Fig. 10. Effective thermal resistances for JSM and JPM heat sinks under (a) $Re_j = 244 \sim 732$, and (b) $\Delta T_{sub} = 20°C \sim 60°C$.

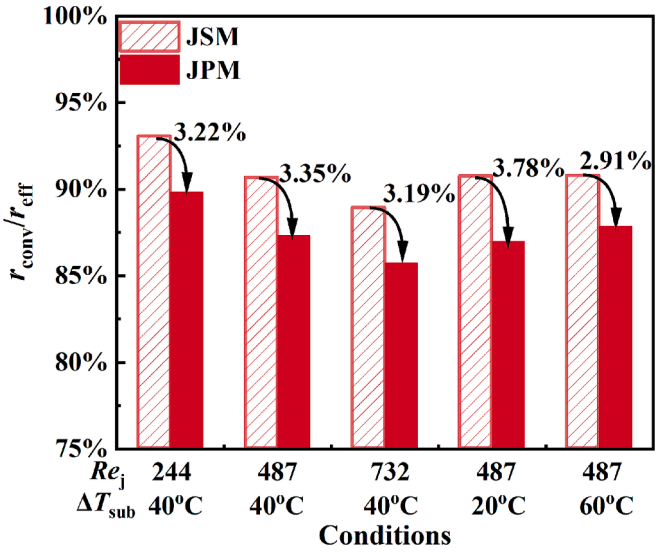


Fig. 11. Ratios of convective thermal resistances for JSM and JPM heat sinks under different conditions.

variations of seven base temperatures and two fluid pressures (i.e. inlet and outlet pressures) at the boiling midterm stage for JSM and JPM, under $Re_j = 487$ and $\Delta T_{sub} = 40°C$. It is found that compared with JSM (Fig. 14(a)), the average oscillation ranges of seven base temperatures and two pressures for JPM (Fig. 14(b)) are decreased by 58.3 % (from 7.41 °C to 3.09 °C) and 62.7 % (from 1.69 kPa to 0.63 kPa), respectively.

Temperature uniformity is usually characterized by employing the standard deviations of temperatures. Generally, the temperature distribution is more uniform when the standard deviation is smaller. The standard deviations of seven base temperatures (defined as Eq. (11)) for JSM and JPM heat sinks under different conditions are presented in Fig. 15. In the single-phase regime, the standard deviation increases linearly with increasing heat flux, reaching its local maximum at ONB. After ONB, the standard deviation decreases suddenly due to the transition from single-phase flow to two-phase flow. The standard deviation in the two-phase regime decreases with increasing heat flux due to the increase in phase-change region. As the heat flux continues to rise, the standard deviation increases again due to the appearance of local dry-out. It also can be seen that the addition of pin–fin structures in the microchannel significantly improves the temperature uniformity for JSM. In the single-phase regime, the standard deviations for JPM are generally lower than those for JSM because the pin–fin structures in the microchannel can enhance flow disturbance. In the two-phase regime, the temperature uniformity for JSM is significantly improved by adding pin-fins, which increase the number of nucleation sites and promote uniform distribution of different phases across the flow region, as discussed above. In comparison to JSM, the average standard deviations of base temperatures after ONB for JPM decrease by 16.0 %~22.3 % under different conditions. The minimum standard deviation of base temperatures for JPM in the two-phase regime decreases to 1.90°C under $Re_j = 487$ and $\Delta T_{sub} = 20°C$.

3.4. Pressure drop and thermal–hydraulic performance

The variation in pressure drops through the distributed jet/microchannel heat sinks with effective heat fluxes for JSM and JPM heat sinks under different conditions are presented in Fig. 16. It is shown that in the single-phase regime, the pressure drop marginally reduces as the heat

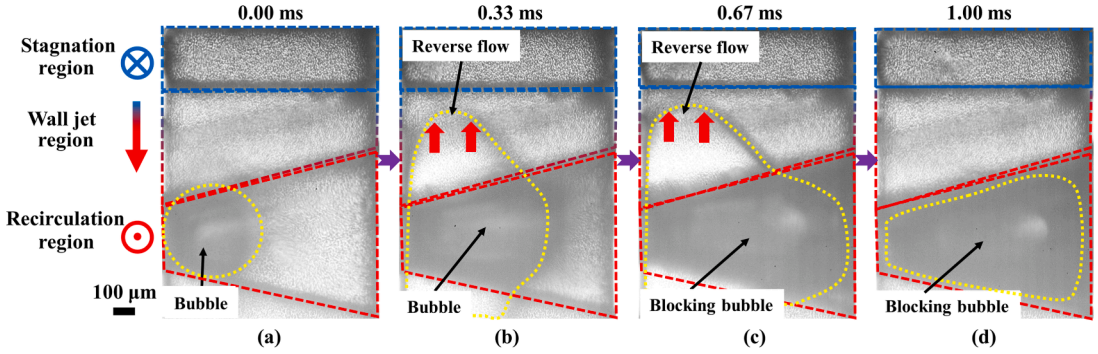


Fig. 12. Evolution of flow patterns at the middle boiling stage for JSM under $Re_j = 487$, $\Delta T_{sub} = 40°C$, and $q_{eff} = 591$ W/cm^2.

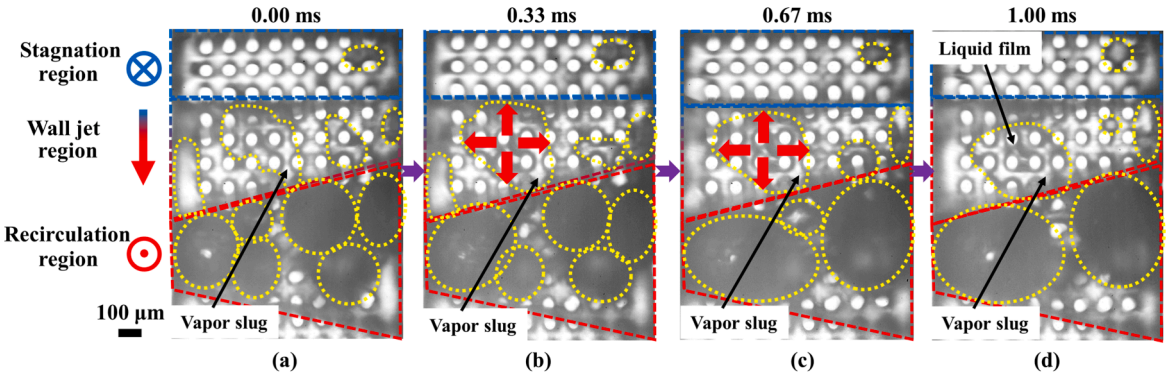


Fig. 13. Evolution of flow patterns at the middle boiling stage for JPM under Re_j = 487, ΔT_{sub} = 40°C, and q_{eff} = 780 W/cm^2.

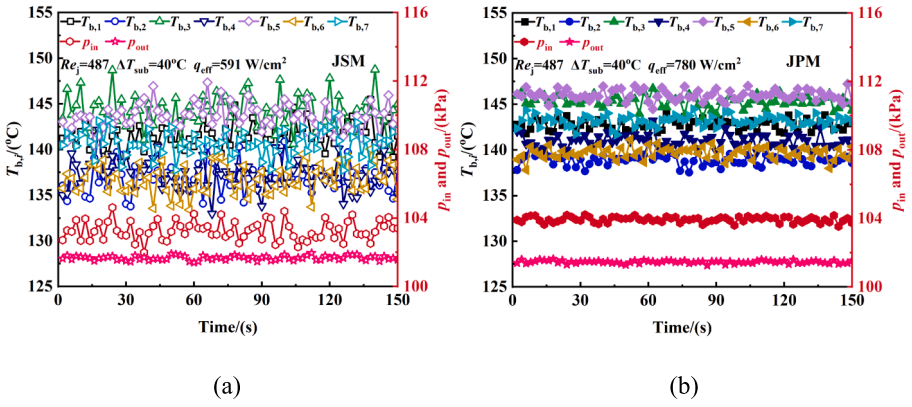


Fig. 14. Transient variations of base temperatures and fluid pressures at the boiling midterm stage under Re_j = 487 and ΔT_{sub} = 40°C for (a) JSM (q_{eff} = 591 W/cm^2), and (b) JPM (q_{eff} = 780 W/cm^2).

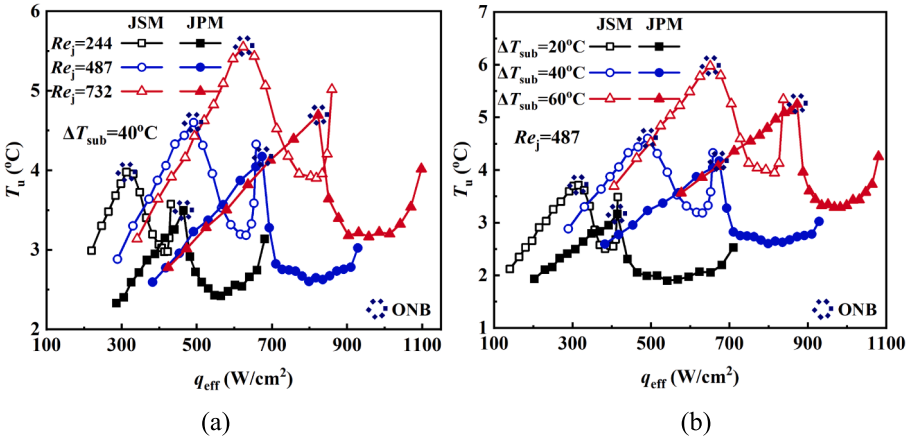


Fig. 15. Standard deviations of base temperatures for JSM and JPM heat sinks under (a) Re_j = 244 ~ 732, and (b) ΔT_{sub} = 20°C ~ 60°C.

flux increases because the increase of fluid temperature causes the decrease of fluid dynamic viscosity. When boiling occurs, the pressure drops increase monotonically with increasing heat flux because of the rising vapor quality in the microchannel. Particularly, an abrupt increase in pressure drop appears after ONB due to the influence of boiling hysteresis (introduced in Section 3.1). The pressure drops for JPM are larger than those for JSM under the same condition due to the increase of channel velocity in the pin–fin microchannel. In addition, the slopes of pressure drop curves for JPM are steeper than JSM in the two-phase regime. This is because the pressure drop largely depends on the vaporization rate of fluid in the two-phase regime, and JPM can produce more nucleation sites and have a higher vaporization rate. Note that the maximum pressure drop for JPM is very small, only 4.20 kPa under Re_j = 732 and ΔT_{sub} = 40°C. It is also observed that the pressure drops increase with increasing jet Reynolds number (Fig. 16(a)). Nevertheless, as seen in Fig. 16(b), the pressure drops in the single-phase regime increase slightly with increasing inlet subcooling because the fluid dynamic viscosity increases with decreasing inlet temperature. The curves of pressure drop in the two-phase regime shift to the right with increasing inlet subcooling due to the boiling occurring at higher heat flux.

As discussed above, the heat transfer performance for JPM is

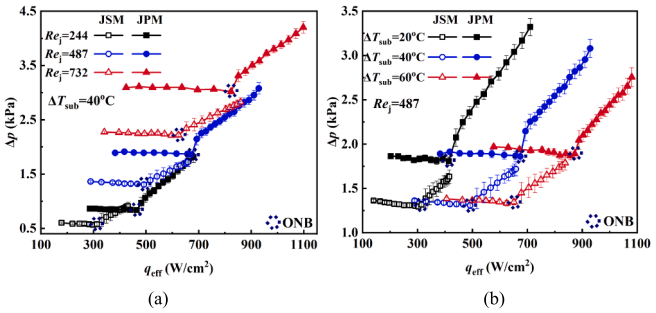


Fig. 16. Pressure drops for JSM and JPM heat sinks under (a) $Re_j = 244 \sim 732$, and (b) $\Delta T_{sub} = 20°C \sim 60°C$.

enhanced at the cost of the increase in pressure drop as compared to JSM. Thus, the performance evaluation criterion (*PEC*, defined as Eq. (13)) is used to comprehensively evaluate the thermal–hydraulic performance of the heat sink. The average *PEC*s for JPM as compared to JSM under different conditions are shown in Fig. 17. It is seen that average *PEC*s for JPM are 1.18 ~ 1.29, greater than 1 for all conditions, which means that JPM has better comprehensive thermal–hydraulic performance than JSM. Moreover, the influence of jet Reynolds number and inlet subcooling on *PEC* for JPM is not obvious in the present work. Table 2 further compares the flow boiling heat transfer performances including CHF, HTC (calculated by Eq. (6)), pressure drop, and *PEC* (compared to the referenced heat sink of JSM), by present work and previous studies on two-phase embedded cooling [11–13,18]. It shows that the hybrid jet/microchannel boiling combining distributed jet with pin–fin microchannel has significant advantages in achieving higher CHF, lower pressure drop, and better comprehensive thermal–hydraulic performance than microchannel boiling.

4. Conclusions

In this study, a hybrid distributed jet/pin–fin microchannel (JPM) heat sink with flow boiling visualization has been constructed successfully in the silicon base. The flow boiling experiments of deionized water in JPM heat sink with jet Reynolds numbers of 244 ~ 732 and inlet subcoolings of 20°C ~ 60°C are conducted and compared with those in distributed jet/smooth microchannel (JSM) heat sink fabricated in the silicon base. Two-phase flow patterns in JPM and JSM are

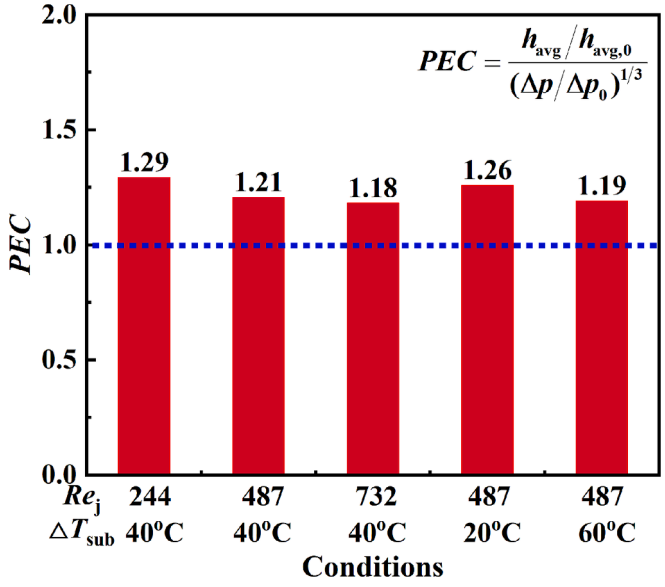


Fig. 17. *PEC*s for JPM as compared to JSM under different conditions.

Table 2
Comparisons of flow boiling heat transfer performances by present work and previous studies on two-phase embedded cooling [11–13,18].

Refs.	Cooling scheme	CHF /(W/cm^2)	HTC/ (kW/(m^2·K))	Pressure drop/ (kPa)	*PEC*
Ma et al. [11]	Wavy microchannel	67.2	13.7	230	0.031
Luo et al. [12]	Microporous microchannel	473	45.0	>27.0	0.165
Cheng and Wu [13]	Interconnected microchannel	428	60.2	88.4	0.183
Drummond et al. [18]	Manifold microchannel	1020	155	120	0.466
Present work	Hybrid distributed jet/smooth microchannel	860	104	2.83	1.00
	Hybrid distributed jet/pin–fin microchannel	**1098**	137	**4.20**	**1.29**

simultaneously captured by a high-speed microscope camera. The flow boiling curve, heat transfer coefficient and effective thermal resistance, boiling stability and temperature uniformity, pressure drop and thermal–hydraulic performance are comprehensively examined and discussed. The following conclusions can be drawn from the present work:

(1) Compared with JSM, JPM can pre-trigger ONB because the pin–fin structures in JPM can produce more nucleation sites than the smooth surface in JSM. CHFs for JPM are significantly enhanced by 27.7 %~70.8 % under different conditions due to the effective prevention of reverse flow and local dry-out, which easily occur in JSM. Specifically, the JPM heat sink achieves an extremely high CHF of 1098 W/cm^2 at a small pressure drop of 4.2 kPa under $Re_j = 732$ and $\Delta T_{sub} = 40°C$.
(2) Compared with JSM, JPM can increase the heat transfer coefficient by 33.4 %~51.6 % and decrease the effective thermal resistance by 22.8 %~32.1 % due to more nucleation sites and larger heat transfer surfaces existing in JPM than in JSM. In particular, the maximum HTC and the minimum effective thermal resistance reach 137 kW/(m^2·K) and 8.7 × 10^{-6} K·m^2/W, respectively, under $Re_j = 487$ and $\Delta T_{sub} = 20°C$.
(3) Better flow boiling stability and base temperature uniformity (in both single-phase and two-phase regimes) are obtained for JPM as compared to JSM. This is because the pin–fin structures in JPM can enhance flow disturbance, promote uniform distribution of different phases, and prevent the reverse flow of rapidly expanding bubbles.
(4) Although the enhancement in heat transfer is at the cost of the increase in pressure drop (due to the increase of flow velocity in the pin–fin microchannel), JPM has superior comprehensive thermal–hydraulic performance than JSM, with *PEC*s for JPM compared to JSM being 1.18 ~ 1.29 under different conditions.

CRediT authorship contribution statement

Jinya Liu: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Huiying Wu:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization. **Xia Hua:** Writing – review & editing, Validation, Investigation. **Jiru Wei:** Writing – review & editing, Validation. **Zhenyu Liu:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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Data availability

Data will be made available on request.

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